

Project AlpineBridge: A Governed AI-Assisted Platform for SME Credit File Preparation at ZKB

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To: Head of Innovation, Zürcher Kantonalbank (ZKB)

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Project AlpineBridge should proceed only as a controlled AI-assisted SME credit-file preparation and workflow project. The customer-visible component, CreditBridge, should improve transparency and reduce friction for SME applicants, while the internal AlpineBridge platform should support governed ingestion, validation and workflow routing. The project should not be positioned as an autonomous credit-decision system. A controlled pilot is recommended if credit-risk co-ownership, data-protection readiness, integration feasibility, operational support and measurable KPI baselines can be secured.

1. Project Description

Scenario and Market Environment

Swiss bank ZKB wants to position itself as modern and forward-thinking without weakening the risk discipline that makes banking trustworthy. Project AlpineBridge addresses this tension through a customer-visible SME credit-file preparation portal, branded as CreditBridge, and a controlled internal AI-assisted ingestion and validation layer, branded as AlpineBridge.

The market problem is practical rather than futuristic. Established micro-SMEs and small SMEs often need liquidity quickly, but standard credit applications still involve PDF statements, tax documents, bank statements, accounting exports, KYC checks, missing-document loops and repeated manual re-keying. For small-ticket cases, the administrative effort can become disproportionate to the expected margin. This creates an opportunity for ZKB to improve customer experience and internal efficiency at the same time (SECO, 2023; Swiss Bankers Association / IFZ, 2025).

The MVP should target established Swiss micro-SMEs and small SMEs applying for standard working-capital loans, overdraft facilities or simple equipment-financing products. Startups without stable financial history, restructuring cases, cross-border groups and complex corporate structures should remain outside the first release. This scope discipline is important because the project is meant to prove a controlled, auditable workflow before any broader rollout.

All business figures in this summary are working assumptions for the project scenario, not internal ZKB data. Suggested baseline assumptions for the pilot are: five to ten working days to complete a standard SME credit file, two to four hours of manual handling per case, frequent bottlenecks around missing documents, manual PDF extraction, accounting reconciliation and KYC checks, and a controlled pilot size of roughly 300 to 500 SME applications. The first business target should be a measurable reduction in manual re-keying and missing-document loops, not an automated credit approval rate.

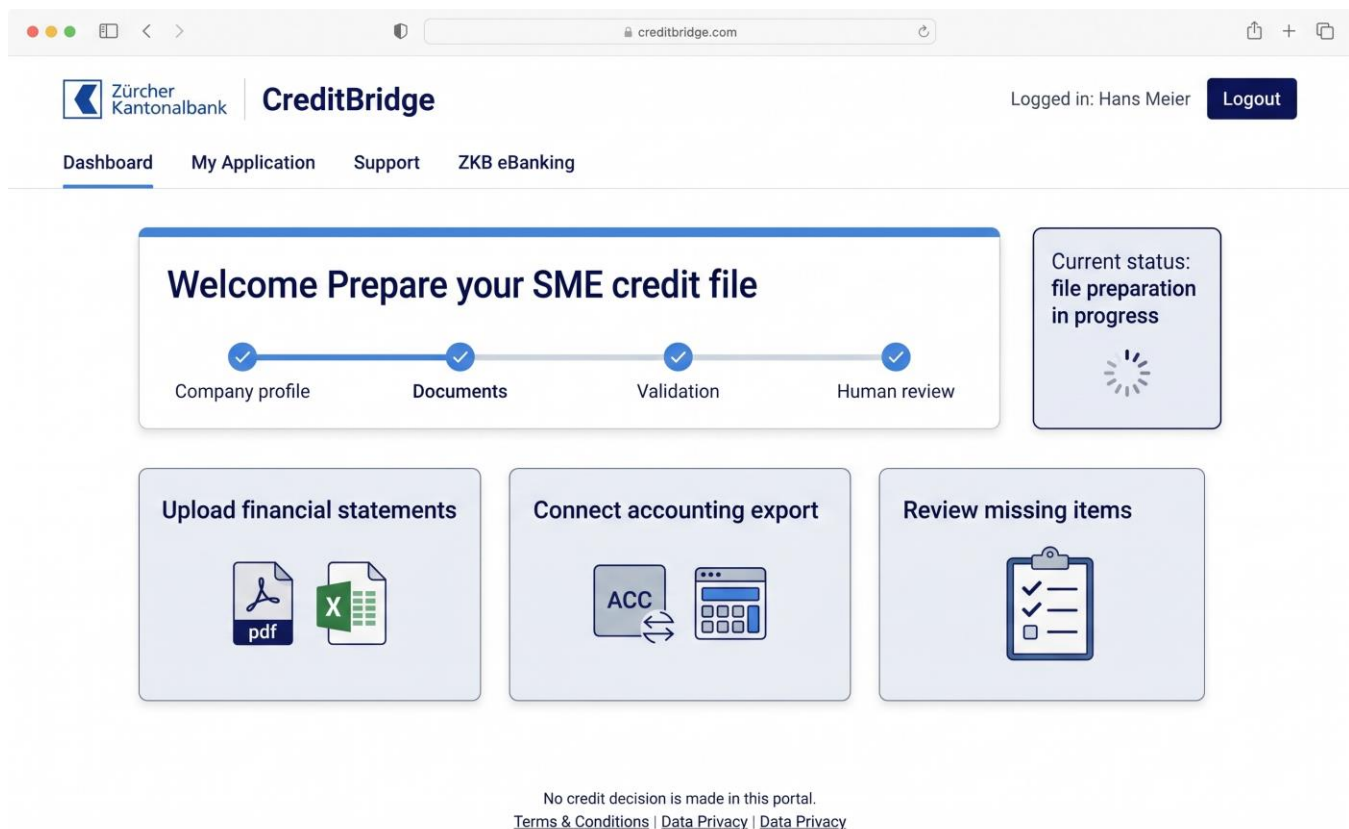


Figure 1: CreditBridge customer portal for SME credit-file preparation.

Controlled Product Scope

CreditBridge is the visible product that potential SME customers interact with. It guides applicants through the preparation of a complete credit file, explains missing documents, allows accounting exports where technically and legally appropriate, and shows whether the file is ready for human credit review.

AlpineBridge is the internal governed layer behind this product. It may use OCR, layout-aware extraction, document classification, accounting-data mapping, anomaly detection, confidence scoring and workflow routing. It must not approve, reject, recommend, rank customers for approval, communicate approval probability, or generate customer-facing credit explanations. Final credit decisions remain in existing ZKB credit processes with accountable human ownership (European Data Protection Board, 2018; European Union, 2024).

The MVP should use a conservative hybrid architecture: OCR and layout-aware extraction for PDFs, deterministic validation rules for accounting consistency, mapping rules for conversion into a Unified Chart of Accounts (UCoA), and statistical confidence scoring for uncertain fields. Generative AI, if used at all, should be restricted to internal summarisation or classification support and must not generate credit recommendations or approval language.

The screenshot shows a web browser window with the URL creditbridge.com. The header features the Zürcher Kantonalbank logo and the CreditBridge name, along with 'My Application' and 'Logout' buttons. The main content area is titled 'Start SME credit-file preparation'. On the left, a 'Credit-File Steps' sidebar lists: 1. Company details, 2. Documents (highlighted), 3. Validation, and 4. Human review. The central area is for uploading documents, with the text 'Upload financial statements, tax documents, bank statements' and buttons for PDF, XLSX, and CSV. Below these is an icon of a document being uploaded to a cloud. On the right, an 'Optional accounting export' section contains buttons for 'Bexio export' and 'Abacus export', and a note about 'Optional data connection - separate authorization required'. A light blue bar at the bottom of the main area states 'File not yet complete'.

Figure 2: CreditBridge application intake with document upload and optional accounting export.

Motivation and Goals

The motivation is to make ZKB visibly easier to use for SMEs while strengthening internal process quality. The customer-facing benefit is a clearer, more transparent application journey. The internal benefit is less repetitive document chasing, fewer manual transcription errors and better-prepared credit files before they reach the responsible credit officer.

The pilot should be judged by measurable operational KPIs: reduction in time-to-complete-file, reduction in manual re-keying, extraction correction rate below agreed thresholds, percentage of cases requiring manual exception handling, SME application abandonment rate, credit-officer acceptance and number of production incidents. The strongest business case is not 'AI decides faster', but 'humans receive cleaner files faster'.

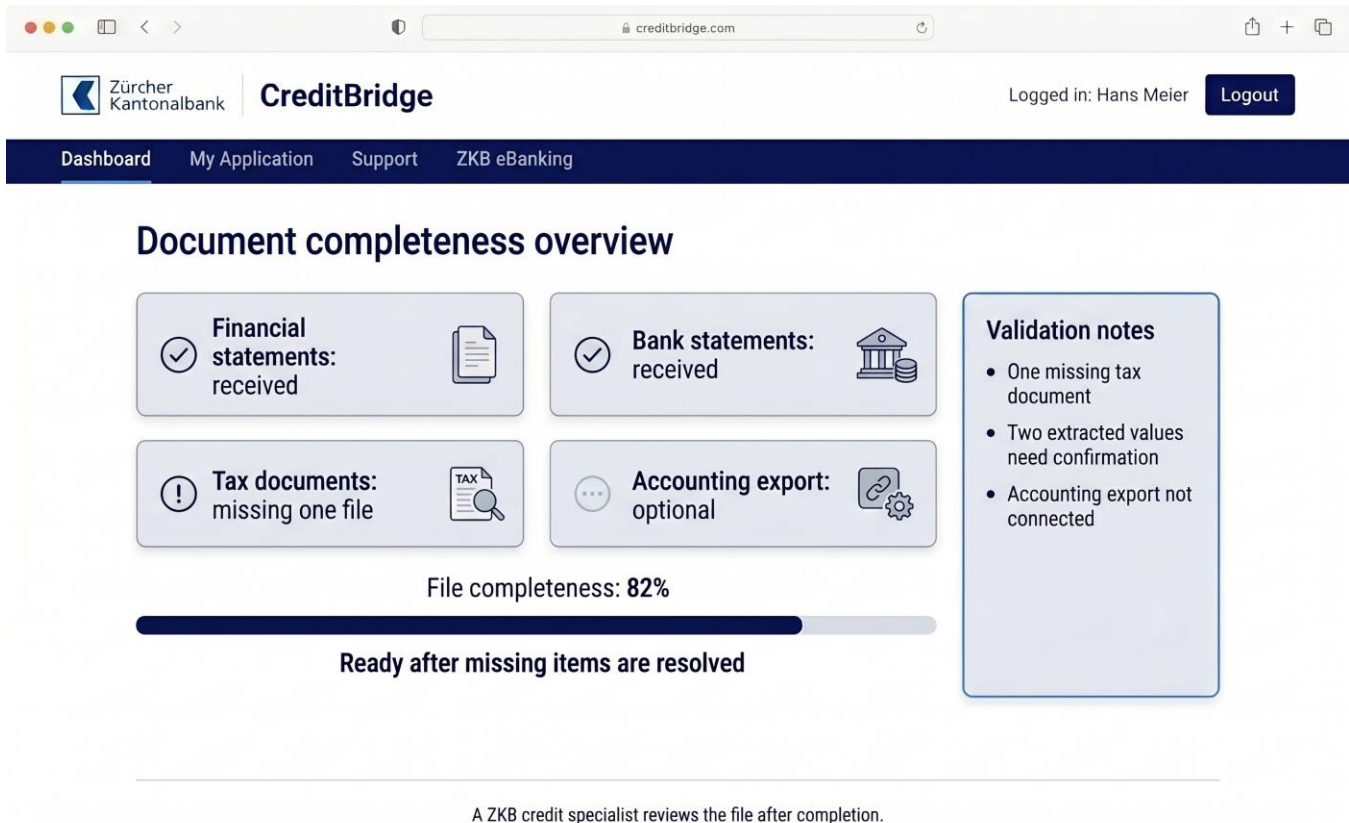


Figure 3: CreditBridge document completeness and validation overview.

Distinctive Features and Crunchpoints

The distinctive feature of the project is the combination of a simple SME-facing product with a complex banking-grade backend. From the customer perspective, CreditBridge should feel like a guided preparation portal. From the bank perspective, AlpineBridge is a large integration and governance project touching identity, document management, KYC, credit workflow, data protection, model monitoring, security operations and auditability.

The first crunchpoint is the boundary between innovation pressure and credit accountability. The sponsor wants a visible AI product, but the safe and credible version must avoid AI credit decisioning. The second crunchpoint is SME simplicity versus banking security: authentication, consent, file upload and accounting export must be secure without becoming so heavy that SMEs abandon the process. The third crunchpoint is human review versus automation ROI: if too many fields need correction, the business case collapses. The fourth crunchpoint is auditability versus data deletion: traceability is necessary, but logs and retained documents must respect retention and deletion concepts. The fifth crunchpoint is vendor speed versus bank control, because external OCR, AI or connector services can accelerate delivery while creating data-location, audit, subcontractor and exit risks.

Project Extent: >300 Person-Months

AlpineBridge is estimated at 340 person-months over 24 months with a budget envelope of CHF 9.5 million. This scale is justified only if the project is understood as a bank-wide integration, governance, security and operational-readiness programme, not as a simple frontend application.

The budget assumption implies roughly CHF 27,900 per person-month when internal staff, external specialists, infrastructure, security testing, audit support, tooling, environments and contingency are included. The team does not work at full capacity for all 24 months. Staffing increases during prototype and pilot work and decreases during hardening and rollout decision activities.

Phase	Duration	Average team size	Effort
Discovery	6 months	8 FTE	48 PM
Prototype	6 months	14 FTE	84 PM
Pilot	6 months	16 FTE	96 PM
Hardening	6 months	14 FTE	84 PM
Steering / audit/ contingency	Across project	-	28 PM
Total	24 months	-	340 PM

Table 1: Delivery scale assumption

The core team should include project leadership, credit-domain ownership, AI and data engineering, core banking integration, enterprise architecture, security, legal/compliance, QA, operations and change management. The important point is not the absolute headcount in one month, but the coordinated availability of senior banking expertise and technical delivery capacity across the full lifecycle.

Promoters and Stakeholder Goals

The primary promoter is the Head of Innovation, whose goal is to make ZKB visibly more modern while creating a reusable AI-assisted capability. The CFO supports the project if it lowers manual effort and improves the cost-to-income logic of small SME lending. The CEO and Board support it only if the reputational risk remains controlled. The Head of Corporate Credit Risk and the CRO should not be treated as opponents; they must become co-owners of the credit-process boundary, validation rules and go/no-go criteria. The CIO, CISO, DPO, Legal, Compliance, Internal Audit and Operations support the project only if it can be embedded into normal bank governance instead of becoming a parallel innovation system.

SME customers are also promoters only if the product genuinely reduces friction. If CreditBridge feels more intrusive, slower or less clear than sending files to a relationship manager, the visible innovation fails even if the backend works.

2. Risks of this Project

The following risks are ranked by their potential to stop the project or destroy its credibility. Each risk is assigned to one primary category: technical issues, project management issues, company issues, or public/legal issues.

Rank 1 - Public/legal issues: AI decision-boundary risk

The most dangerous risk is that AlpineBridge is presented, designed or perceived as an AI credit-decision system. This would contradict the safer project logic and expose the bank to stronger explainability, accountability and reputational risks. The cause is a conflict between innovation marketing and banking control. The mitigation is to prohibit approval scores, eligibility labels, automated recommendations and customer-facing AI explanations in the MVP. The system may show completeness, confidence and validation status only. The trigger for escalation is any roadmap item or mockup that suggests 'approved', 'eligible', 'credit score' or 'AI recommendation'.

Rank 2 - Public/legal issues: data protection and banking secrecy risk

SME files may contain personal data of owners, directors, guarantors, employees and beneficial owners, as well as sensitive financial information. The project must distinguish between data required for pre-contractual credit processing, data required by legal or regulatory obligations and optional data such as accounting-software exports. Optional data connections require separate transparent authorization and must not become a hidden condition for file submission. Required controls include a data-flow map, DPIA, purpose limitation, role-based access, encryption, processor agreements, deletion rules, breach response and a retention concept for audit logs (Federal Assembly of the Swiss Confederation, 2020).

Rank 3 - Technical issues: legacy integration risk

The largest technical risk is integration with identity, customer master data, document management, KYC, workflow and credit-risk systems. Banking systems contain old interfaces, strict release windows and limited tolerance for unstable data. Mitigation requires early architecture sign-off, API contracts, test environments, rollback planning, batch fallback and clear ownership for each integration.

Rank 4 - Technical/company issues: model-risk, bias and uneven data-quality risk

Even without automated credit decisions, extraction and validation can create unfair friction. SMEs with non-standard documents, smaller accounting systems, mixed personal/business records or different language/canton formats may receive more manual flags. Mitigation requires validation sets by document type, accounting software, legal form, company size, industry and language, plus monitoring of correction rates and manual-review rates. AI uncertainty must never create a negative credit consequence by itself (NIST, 2023; ISO, 2023).

Rank 5 - Technical issues: data-quality and extraction-error risk

OCR and layout-aware extraction may misread amounts, dates, legal names or balance-sheet line items. A wrong extracted number can damage file preparation even if no decision is automated. Mitigation requires confidence scoring, human review of low-confidence fields, reconciliation rules, four-eye checks for critical values and audit trails of every correction.

Rank 6 - Company issues: business-case and ROI risk

The project may not reduce manual effort enough to justify the budget. If extraction correction rates are high, if SME adoption is low or if credit officers still re-check everything manually, AlpineBridge becomes an expensive workflow layer with limited value. Mitigation requires baseline measurement before the pilot and hard KPI gates on manual handling time, time-to-complete-file, cost per processed case, correction rate, abandonment rate and officer acceptance.

Rank 7 - Company/project management issues: SME onboarding and adoption friction

CreditBridge is customer-visible, so adoption is a central risk. If SMEs must complete complex authentication, grant unclear accounting access, upload too many files or interpret technical validation warnings, the portal may feel harder than the current relationship-manager channel. Mitigation requires user testing with real SMEs, plain-language consent, short application paths, manual alternatives and abandonment monitoring.

Rank 8 - Company issues: credit-department resistance

Credit officers and relationship managers may resist the platform if they believe it replaces judgment or monitors their work. Mitigation is to make them co-designers, not late recipients. The system should remove repetitive document chasing, not professional assessment. A trigger is the appearance of manual shadow processes or refusal to trust validated inputs.

Rank 9 - Project management issues: scope creep

The project can expand from credit-file preparation into AI advisory, autonomous decisioning, continuous monitoring, market intelligence and portfolio analytics before the MVP works. Mitigation requires a written out-of-scope list, steering approval for scope changes, separate phase gates and the rule that the first production release must solve file preparation only.

Rank 10 - Technical issues: cybersecurity, malicious-document and vendor risk

The portal is exposed to uploaded financial documents and possible accounting connectors. Threats include data theft, account takeover, malicious PDFs, hidden prompt-injection content, insecure output handling, compromised vendors and unreliable APIs. Mitigation requires malware scanning, content isolation, least-privilege access, security testing, OWASP-style controls for AI components, vendor audit rights, data-location restrictions, exit plans and operational monitoring (OWASP Foundation, 2024).

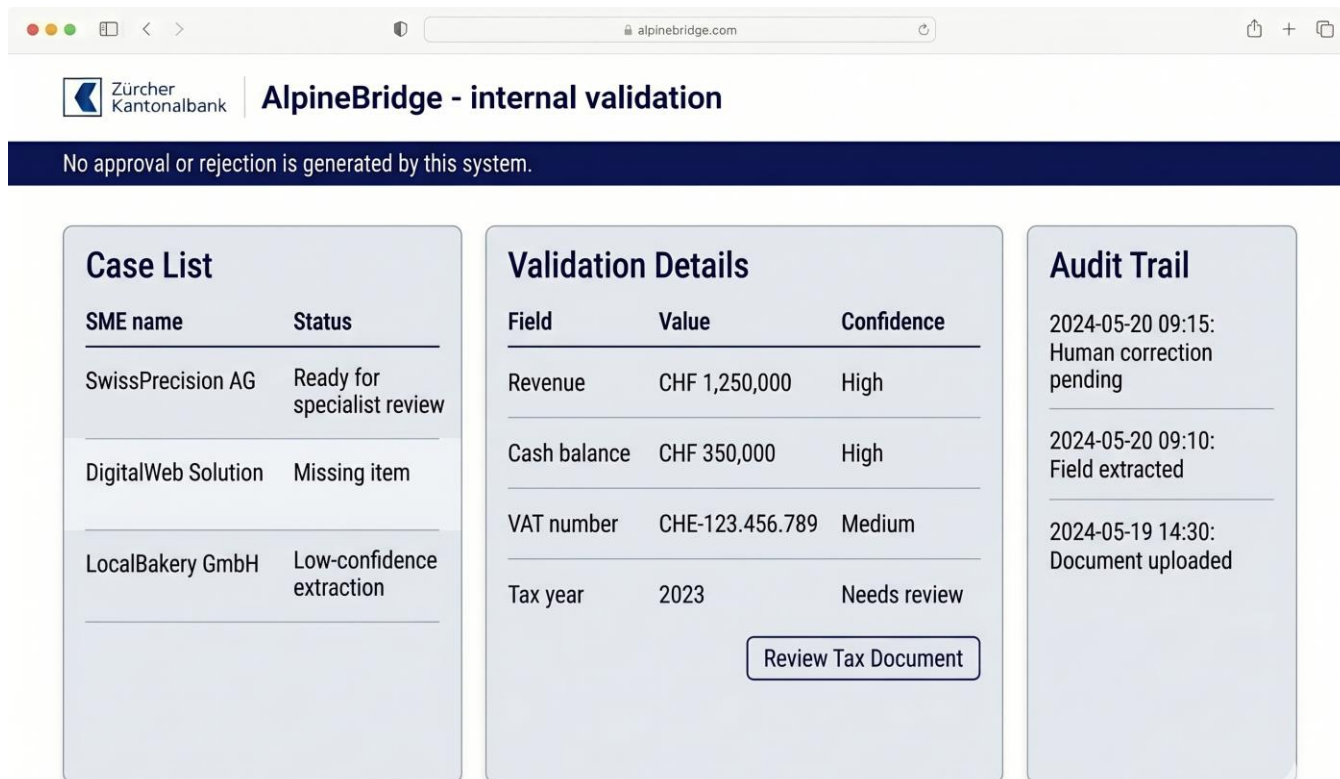


Figure 4: AlpineBridge internal validation screen for human review.

3. Peopleware

AlpineBridge can fail because of peopleware even if the software works. The project touches status, accountability, professional judgment and control functions. Therefore, formal roles and informal power must be managed deliberately.

Formal and Informal Roles and Acting Parties

Actor group	Person	Role
Head of Innovation	M.A.	Sponsor
CEO / Board / CFO	J.E.	Conditional
Credit Risk / CRO	T.H.	Gatekeeper
CIO / Architecture	C.L.	Cautious
CISO / DPO / Legal	R.S.	Control
Internal Audit	C.B., G.M.	Assurance
Credit officers	B.B., C.P.	Adoption
Operations / Service desk	K.A., T.P., R.G., Y.A.	Run
AI / data / vendors	M.L., J.L., T.T., G.E.	Build
SME users	S.T., L.B., U.F.	External test

Table 2: Simplified stakeholder map

The Head of Innovation is the visible sponsor. Her value is strategic visibility and access to senior management. Her risk is over-selling the system as disruptive AI. She must protect the boundary between a modern customer journey and non-automated credit decisioning.

The CEO, Board and CFO are not daily project members, but they shape the political environment. They should receive short steering updates focused on KPI evidence, cost discipline, reputational safety and go/no-go criteria rather than technology excitement.

Corporate Credit Risk and the CRO are the most important gatekeepers. They must co-own validation rules, exception handling, acceptance criteria and the decision boundary. If they are treated as late reviewers, they can block production even after successful technical delivery.

The CIO and Enterprise Architecture function decide whether the solution can enter the real IT landscape. They need enough authority to reject unsafe shortcuts and enough pragmatism to allow controlled experimentation in the pilot.

The CISO, DPO, Legal, Compliance and Internal Audit functions define the conditions for production. Their influence should appear early through data-flow documentation, DPIA, processor review, access control, penetration testing, audit logging and retention concepts. They should not be used as end-stage approvers only.

Credit officers and relationship managers are daily users and informal adoption controllers. If they feel replaced, they will create workarounds. If they see that the system reduces document chasing and improves file quality, they can become the strongest practical promoters.

Operations and the service desk must be involved before the pilot. A system that works in a demo can fail in production if no one feels responsible for incidents, connector failures, access problems, monitoring alarms or customer support questions.

AI engineers, data engineers and external vendors provide the technical capability. They should not define the credit process alone. Their work must be paired with experienced banking staff so that the system becomes operationally usable, not just technically impressive.

SME users are not passive test participants. They are the external adoption judges. If the process feels intrusive, confusing or slower than the current relationship-manager channel, the customer-visible innovation fails. SME associations and pilot customers can therefore provide useful early signals on trust and usability.

The project also has a reputational stakeholder: public perception. A customer-facing AI credit-preparation portal can easily be misunderstood as 'AI decides SME loans'. Communication must therefore avoid exaggeration and explain that the system prepares files for accountable human review.

Required Personalities and Possible Conflicts

AlpineBridge requires senior, high-standing personalities rather than only young innovation profiles. The sponsor should have board access but enough discipline not to oversell AI. The credit-risk co-owner should be an experienced, conservative credit professional with strong internal legitimacy. The enterprise architect should be senior enough to refuse unsafe shortcuts. The DPO, CISO, Legal and Compliance representatives require early veto power, not only review power at the end. The product owner should understand daily SME credit operations, not only digital channels.

The main conflict is between speed and control. Innovation teams may want visible results quickly, while banking operations require specifications, test data, release windows, rollback plans and change approval. Another conflict is between automation targets and human accountability. The project succeeds only if automation is framed as better preparation for expert work, not as replacement of expert work.

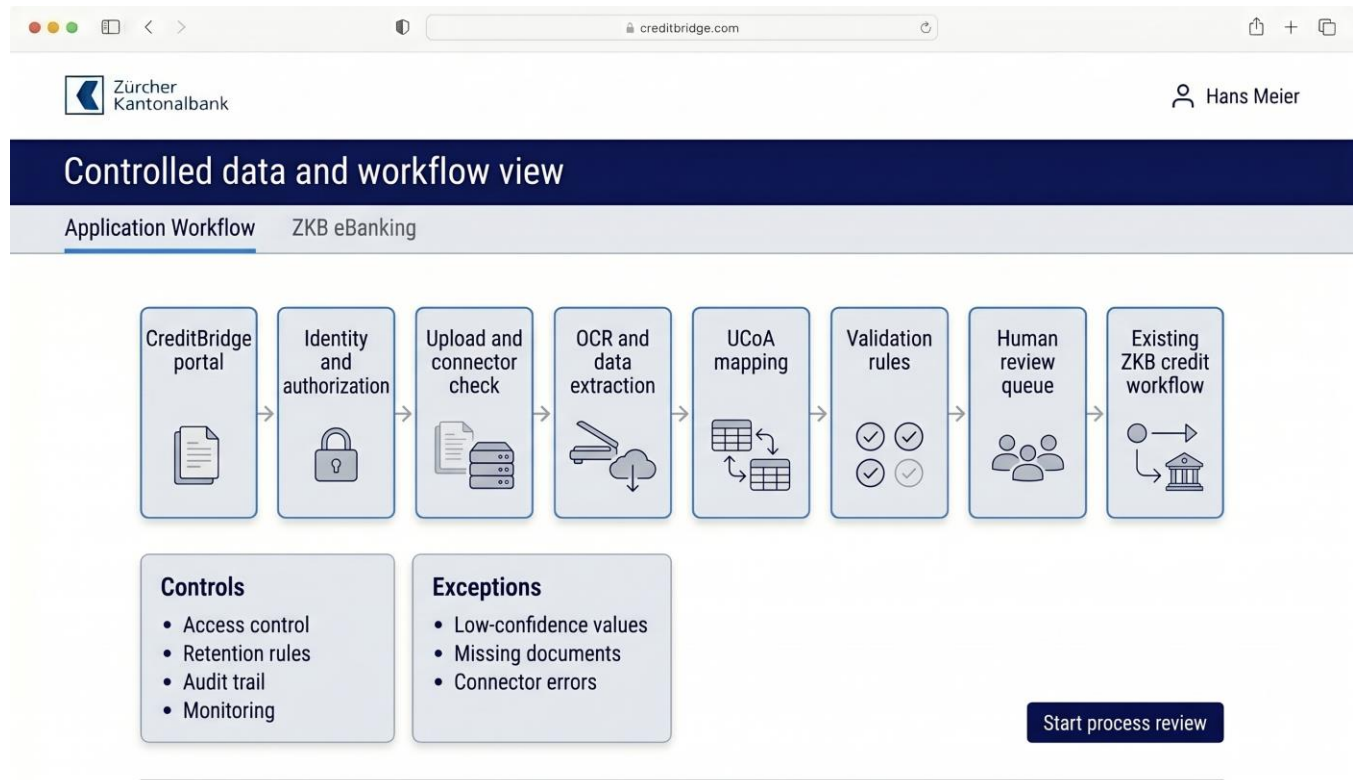
4. Embedment into ZKB's IT Strategy

AlpineBridge fits ZKB's IT strategy only if it is positioned as controlled intelligent middleware. It must not become a parallel credit system. Its strategic value lies in creating a reusable, governed ingestion and validation capability that connects modern customer interfaces with stable backend systems.

Role and Position in the IT Landscape: Intelligent Middleware

The SME-facing CreditBridge frontend collects documents and optional accounting exports. AlpineBridge validates inputs, maps them to the UCoA, records confidence scores and routes exceptions to humans. Only validated structured data and transparent metadata are handed to existing credit workflow and risk systems. This protects the bank from creating an uncontrolled shadow system.

The architecture should be two-speed only in a controlled sense. The frontend and extraction layer may iterate faster during the pilot, but production integration, access control, retention, monitoring, cyber controls and auditability must follow normal banking governance.



AlpineBridge prepares structured inputs; credit decisions remain in approved bank processes.

Figure 5: AlpineBridge controlled workflow from SME portal to existing credit systems.

Recommended Data and Control Flow

- SME opens CreditBridge and authenticates through approved identity and authorization mechanisms.
- The system collects required documents and, where separate authorization exists, optional accounting system exports.
- Uploaded files pass malware checks and content isolation before OCR and layout-aware extraction.
- Extracted values are mapped to the UCoA and checked against deterministic validation rules.
- Low-confidence fields, inconsistencies and exceptions are routed to credit officers.
- Every extraction, correction, model version, data source and handover is logged for auditability.
- Validated structured data is transferred to existing ZKB credit workflow and risk systems.
- Retention, deletion, breach response and operational monitoring remain part of the production operating model (FINMA, 2022).

Data Protection and Governance Embedment

The data-protection design should be specific to the processed data. SME financial statements are needed for credit-file preparation and require confidence scoring plus deletion rules. Bank statements contain sensitive transaction data and require need-to-know access and encryption. Owner and guarantor data require clear legal-basis documentation and strict role separation. Accounting exports create connector and vendor dependency risks. Audit logs are necessary for traceability but must avoid excessive personal data and must have their own retention concept. AI correction data may be useful for monitoring, but reuse beyond the original purpose must be explicitly controlled.

For a Swiss deployment, FINMA expectations, Swiss banking secrecy, the Swiss Federal Act on Data Protection, operational resilience and internal model/governance standards are primary. GDPR and the EU AI Act are relevant as benchmarks, for EU-related cases, EU providers or future-proof governance, but they should not be presented as the only legal basis of the project (FINMA, 2022; Federal Assembly of the Swiss Confederation, 2020).

Correlation with Business and IT Strategy

The business strategy is to become more attractive to SMEs without weakening risk governance. The IT strategy is to modernize customer interaction while keeping core banking stable, secure and auditable. AlpineBridge supports both strategies if it remains a governed integration capability that improves process quality and customer transparency instead of becoming an uncontrolled AI lending engine.

5. Potential of this Project

The project potential is not the use of AI by itself. Its value lies in creating leverage across economics, technology, organisation and customer experience. These levers should be evaluated early because they define whether AlpineBridge should continue, be reused, pivot or stop after the pilot.

Economic levers

The economic lever is lower handling effort per standard SME credit file. If document chasing, manual re-keying and avoidable correction loops decrease, ZKB can serve small-ticket SME cases more efficiently. This does not mean automatic lending. It means credit specialists spend less time preparing files and more time assessing relevant risks.

Technological levers

The technological lever is a reusable ingestion, extraction, validation and audit capability. If built properly, AlpineBridge can later support annual reviews, covenant checks, internal document classification or other controlled banking workflows. This creates momentum beyond the first SME product.

Organisational levers

The organisational lever is cross-functional learning between innovation, credit risk, IT, compliance, operations and frontline relationship managers. The bank learns how to bring AI-assisted workflows into production without bypassing control functions. This is strategically more valuable than a flashy prototype.

Personal and customer levers

The customer lever is transparency. SMEs should know which documents are missing, which values need confirmation and when the file is ready for human review. The employee lever is relief from repetitive preparation work. Both groups must experience the system as support, not surveillance or replacement.

Strategic options

If pilot KPIs are met and incidents remain low, ZKB can continue by expanding to more SME credit products. If internal efficiency is good but customer adoption is weak, the capability can be reused as a back-office workflow tool. If extraction works but the credit use case remains too sensitive, the project can pivot to annual reviews, covenants or document classification. If data quality, compliance or integration fail, the bank should stop before full production and preserve the architectural learnings.

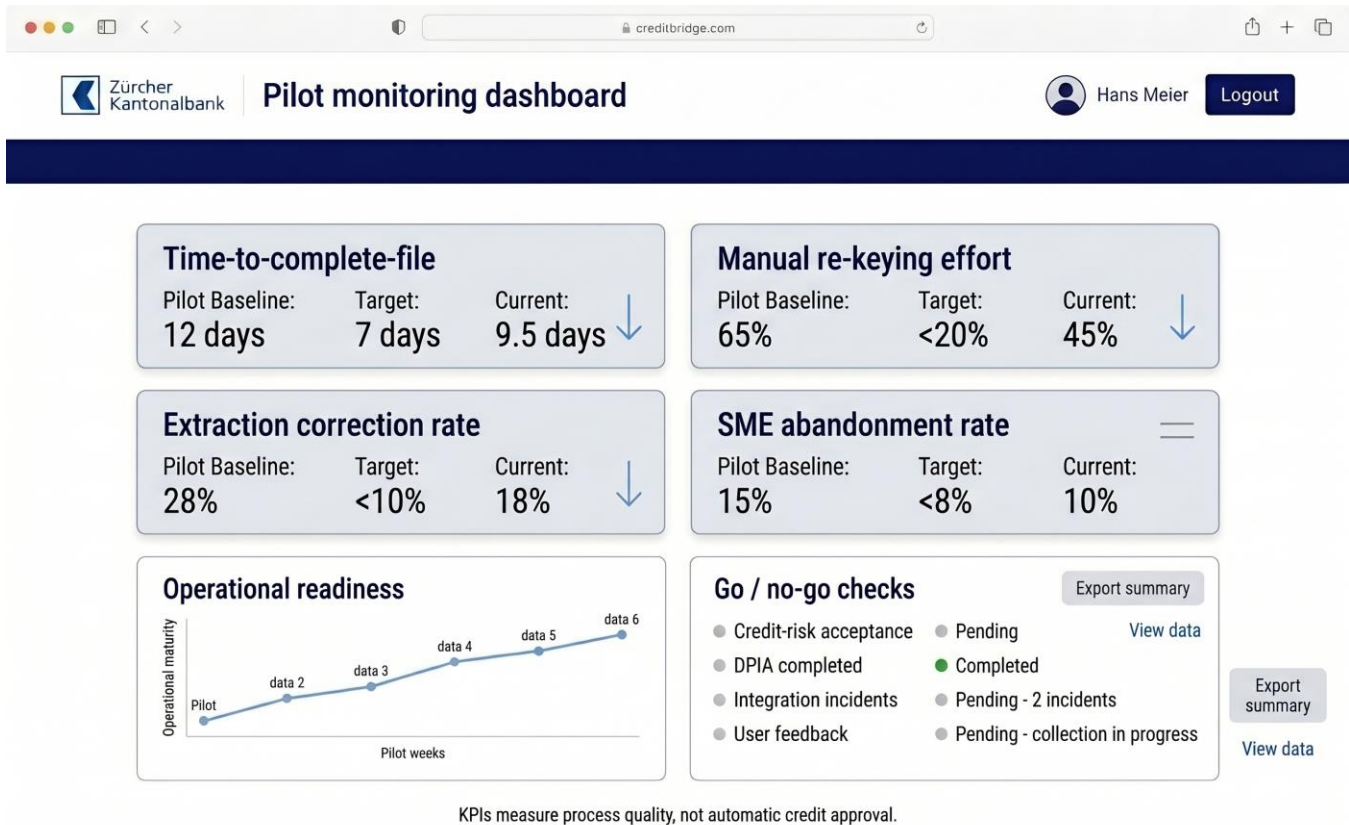


Figure 6: AlpineBridge pilot monitoring dashboard with operational KPIs.

6. Conclusion

Project AlpineBridge should proceed only as a controlled AI-assisted SME credit-file preparation and workflow project. This version is innovative enough to be visible to customers through CreditBridge, but disciplined enough to remain acceptable in a regulated banking environment. The main condition for success is not more AI functionality, but clearer boundaries: no automated credit decisioning, no approval language in the UI, no hidden use of optional data, no late involvement of control functions and no rollout without measurable process improvement.

The project should move forward through a controlled pilot if the bank can secure credit-risk co-ownership, architecture sign-off, data-protection approval, operational support and realistic KPI baselines. It should be stopped or redirected if the AI boundary becomes unclear, if data quality is insufficient, if SME adoption is weak, or if integration and operating-model risks remain unresolved.

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